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Exercise Sheet 5

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Prove that the set of all irrational numbers is uncountable.

Exercise 2.
Prove or disprove: The set $\mathbb{Z} \times \mathbb{Q}$ is countable.

Exercise 3.
Prove that the function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ defined as

$$f(m, n) = 2^{m-1}(2n - 1)$$

is bijective.

Exercise 4.
Prove that if A and B are finite sets with $|A| = |B|$, then any surjection $f : A \to B$ is also an injection. Show that this is not necessarily true if A and B are not finite.

Exercise 5.
Show that $|\mathcal{P}(\mathbb{N} \times \mathbb{N})| = |\mathcal{P}(\mathbb{N})|$

★ **Exercise 6.**
Consider the Cantor triadic set C defined as follows[1]

$$C := \{x \in [0,1] \mid d_i(x) \in \{0,2\}, \forall i \in \mathbb{N}\}$$

Now, the Cantor-Lebesgue function F is defined by

$$F : C \to [0,1], \quad F\left(\sum_{i=1}^{\infty} \frac{a_i}{3^i}\right) := \sum_{i=1}^{\infty} \frac{a_i}{2^{i+1}}$$

(a) Show that $F(0) = 0$ and $F(1) = 1$.

(b) Show that F is well-defined and continuous on C.

(c) Show that F is surjective.

(d) Conclude that C is uncountable.

★ **Exercise 7.**
Prove Theorem 3.16 from the lecture notes.

Theorem (Cantor-Bernstein-Schröder)**.** *Let A and B be sets.*

$$A \preceq B \ \wedge \ B \preceq A \implies A \sim B$$

[1] Taken from Prof. Da Lio's Analysis III, Sheet 5